



Integrating Social Context Into EmoBertGCN for Sentiment Analysis of Filipino Political Discourse

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Background and Problem

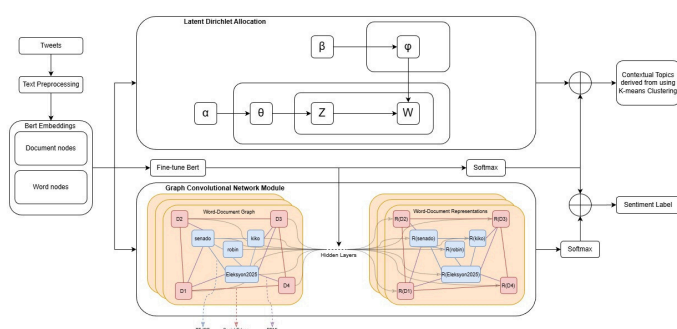
Despite recent advances in sentiment analysis using transformer-based models like BERT and graph neural networks (GNNs), several critical challenges remain for analyzing political discourse in low-resource languages such as Tagalog. Furthermore, olitical datasets are often severely class-imbalanced, with negative sentiments dominating and minority classes being systematically underrepresented, leading to biased and unreliable predictions. Current GNN-integrated frameworks like BertGCN (Lin et al., 2021) and EmoBertGCN (Aquino et al., 2022) capture document-word and word-word relationships but ignore the conversational context between posts under a conversation thread thereby missing the emotional contagion (Hatfield et al., 1993; Coviello et al., 2014) and homophily (McPherson et al., 2001) effects that shape sentiment expression on social media. There is also limited research applying these advanced architectures to Philippine-specific datasets, especially those situated during key electoral periods. This study addresses these gaps by asking: Does incorporating conversational context into a graph-based sentiment analysis framework improve classification performance on imbalanced, low-resource political discourse data?

Research Objectives

This research has two main goals:

- To assess whether incorporating conversational context improves sentiment analysis performance on low-resource, class-imbalanced political discourse datasets.
- To expand NLP resources for Tagalog corpora by producing a labeled dataset based on the 2025 Philippine general midterm elections, supporting future studies on political communication during this key event.

Methodology



- Extends the EmoBertGCN (Aquino et al., 2022) framework by constructing a heterogeneous graph that includes document nodes (tweets) and word nodes, with four types of edges: TF-IDF based document-word connections, PPMI based word-word connections, and newly added social interaction edges that connect documents within the same conversation thread
- BERT embeddings initialize the document nodes, which are then processed through a Graph Convolutional Network (GCN) module to propagate sentiment information across the graph
- The GCN outputs document representations that are combined with the original BERT embeddings via an interpolation parameter ($\lambda = 0.7$), followed by a softmax layer for sentiment classification
- Separately, a topic modeling pipeline using BERT-LDA with UMAP dimensionality reduction and K-means clustering derives contextual topic clusters for interpretability

Discussion of Results

Comparison	Mean A	Mean B	p-value	Sig.
Conversation Aware BertGCN vs. Roberta Baseline	0.8487	0.8534	0.642	No
Conversation Aware BertGCN vs. BertGCN Baseline		0.8456	0.401	No

The study conducted one-tailed permutation tests with Westfall-Young correction to compare the proposed model (BertGCN with conversation edges) against two baselines. The proposed model achieved a macro F1 score of 84.9%, compared to 85.3% for the RoBERTa baseline and 84.6% for the original EmoBertGCN. The comparison against RoBERTa yielded a p-value of 0.642 (Cohen's $d = -0.163$), while the comparison against EmoBertGCN yielded a p-value of 0.401 (Cohen's $d = 0.114$). The null result is attributed to two factors:

- Sparse conversation connectivity in the dataset (only 23.5% of posts were connected via reply threads)**
- A strong RoBERTa baseline fine-tuned specifically for code-switched Tagalog sentiment, which left little room for GCN-driven gains.**



Topic modeling revealed five interpretable discourse clusters from the election-period tweets:

- Cluster 0 (Pro-Opposition / Bam-Kiko Reformist Bloc)** – Predominantly positive sentiment, featuring keywords such as bam, kiko, pangilinan, aquino, salamat, matalino, governance
- Cluster 1 (Anti-Administration / Duterte Loyalist Bloc)** – Negative sentiment dominant, featuring Duterte, Sara, Digong, Bato, Robin, PDP, Quiboloy along with derogatory terms like bobo and tuta
- Cluster 2 (COMELEC & Election Process News)** – Neutral sentiment, focused on factual election updates with keywords such as elections, mayo, proklamasyon, automated, canvassers, radyopilipinas, news5
- Cluster 3 ("Bobotante" / Voter Frustration & Mockery)** – Negative sentiment, mocking voter intelligence rather than endorsing candidates, featuring strong derogatory language
- Cluster 4 (General Public / Undecided Mixed Queries)** – Neutral sentiment, capturing public curiosity and questions about candidates and processes with keywords like sino, pwede, kahit, siguro, iboboto

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